

EXAM

This is a 3 hours exam. No documents allowed.

I – EXAM QUESTIONS (2/3 of the points)

Please discuss the statements, making explicit use of the models and results that we have discussed in this course. You should prepare answers for *two of the three* statements below.

- 1 – “Business cycles are the result of efficient adjustments in economic activity in response to exogenous changes in real economic fundamentals, hence government policy should refrain from active attempts to stabilize economic output.”
- 2 – “Consumption-based explanations of asset price fluctuations are unable to account for the observed fluctuations in equity and bond markets. These fluctuations must therefore be the result of bubbles, bounded rationality, herd behavior or other market inefficiencies.”
- 3 – “The proper design of monetary policy rules should make the coordination on a determinate path for inflation through active interest policy responses its primary objective.”

II – ASSET PRICES (TAKEN FROM DAVIDE DEBORTOLI)(1/3 of the points)

Consider a pure endowment economy, where the representative agent makes his choices regarding consumption (C) and savings (B) solving the following problem:

$$\begin{aligned} \max_{\{C_t, B_t\}_{t=0}^{\infty}} \quad & E_0 \sum_{t=0}^{\infty} \beta^t \frac{C_t^{1-\sigma}}{1-\sigma} \\ \text{s.t.} \quad & Y_t + B_{t-1} \geq C_t + Q_t B_t, \end{aligned}$$

where Q_t is the price of a risk-free discount bond delivering one unit of consumption in period $t+1$. The endowment Y_t is assumed to follow the exogenous process:

$$\frac{Y_t}{Y_{t-1}} = \varepsilon_t,$$

where $\log \varepsilon_t \sim N(0, s^2)$.

- 1 – Derive the first order conditions of the agent’s problem and the market clearing condition.

Denote $\beta^t \lambda_t$ the Lagrange multiplier of the budget constraint in period t . First order condition w.r.t. C_t is $C_t^{-\sigma} \lambda_t$, FOC w.r.t. B_t gives $\lambda_t Q_t = \beta E_t \lambda_{t+1}$.
Good market equilibrium is $Y_t = C_t$ and bond market equilibrium is $B_t = 0$.

- 2 – Define a competitive equilibrium of this economy. Find an expression for the equilibrium value of Q_t that involves β , Y_t , Y_{t+1} , σ and the conditional expectation operator. (call this equation (1))

A competitive equilibrium is a sequence of quantities C_t , B_t and prices Q_t such that (i) FOC and budget constraint of the household are satisfied and (ii) markets clear.

We easily obtain

$$Q_t = \beta E_t \left(\frac{Y_{t+1}}{Y_t} \right)^{-\sigma} \tag{1}$$

- 3 – What is the value of Q in a perfect foresight (deterministic) steady state?

$$Q_t = \beta \tag{2}$$

- 4 – Solve equation (1) to find an exact expression for the bond price Q_t as a function of constants and/or variables that are observable at time t or earlier. (You may need to use the following relationship: if X follows a log-normal distribution and $x = \log X$, then $E(X) = Ee^x = e^{E(x) + \frac{1}{2} \text{Var}(x)}$).

From the Euler equation,

$$Q_t = \beta E_t \left[\exp \left\{ -\sigma \log \underbrace{(Y_{t+1}/Y_t)}_{\log \varepsilon_{t+1}} \right\} \right]$$

Using the property of the log-normal distribution:

$$Q_t = \beta \exp \left\{ -\sigma E_t \left[\underbrace{\log \varepsilon_{t+1}}_{=0} \right] + \frac{\sigma^2}{2} \underbrace{\text{Var}(\log \varepsilon_{t+1})}_{=s^2} \right\}.$$

Therefore

$$Q_t = \beta \exp \left\{ \frac{\sigma^2 s^2}{2} \right\} \quad (3)$$

5 – What is the average value of Q ? Is it the same as the steady state value computed in question **3**? Why?

As seen in equation (3), Q_t is constant, and its average value is therefore $\beta \exp \left\{ \frac{\sigma^2 s^2}{2} \right\}$. When there is risk in the economy ($s > 0$) and agents are risk averse ($\sigma > 0$), the price of the risk fee asset is higher.

6 – How does the variance of the endowment process (s^2) affect the bond price? Why?

Observe equilibrium price equation (3). As the uncertainty s^2 increases, the bond price increases. This comes from precautionary savings motive. As the economy becomes more volatile, households are more willing to buy bonds in order to prepare for the uncertain future. In order to clear the market, the price of the bond should become higher. (Note that $u'''(c) > 0$ is required to have precautionary savings motive and the given utility function satisfies this requirement)

7 – Find the equilibrium solution for the interest rate $i_t = -\log Q_t$.

Take log to the both sides of the equation (3) and multiply by -1 yields

$$i_t = -\log Q_t = -\log \beta - \frac{\sigma^2 s^2}{2}$$

8 – Suppose that neither you nor your computer could find the exact solution obtained in question 4 above. Log-linearize equation (1) around the steady-state, and obtain an approximate solution for the interest rate as a function of constants and/or variables which are observable at time t or earlier. How does the interest rate depend on the variance of the endowment process according to this approximate relation?

Rearranging the Euler equation we get

$$E_t \beta Q_t^{-1} \left(\frac{Y_{t+1}}{Y_t} \right)^{-\sigma} - 1 = 0$$

which can be rewritten as

$$E_t [\beta Q^{-1} \exp \{-\hat{q}_t - \sigma(\hat{y}_{t+1} - \hat{y}_t)\}] - 1 = 0$$

where for a generic variable x_t we have used the definition $\hat{x}_t = \log X_t - \log X$. (note that $\hat{q}_t = 0$). Taking a first-order (Taylor) approximation to the above expression around the steady state, we obtain

$$-\hat{q}_t - \sigma \underbrace{E_t [\hat{y}_{t+1} - \hat{y}_t]}_{=E_t \log \varepsilon_{t+1}=0} = 0;$$

Since $\hat{q}_t = \log Q_t - \log Q = -i_t - \log \beta$, we obtain

$$i_t = -\log \beta \quad (5)$$

Notice that (4) and (5) are different. This comes from the fact that log-linearization loses the information related to the second moments(or above).

9 – Assume that the standard deviation of the endowment process (denoted s) can take values in the interval $[0.01, 0.1]$. Comparing the expressions obtained in questions **7** and **8**, calculate the approximation error (in percentage points), first assuming that the agent is risk-neutral ($\sigma = 0$) and then assuming that the agent has a degree of risk-aversion $\sigma = 5$. Discuss

Given $s \in [.01, .1]$, one has $s^2 \in [.0001, .01]$. There are two cases.

Case 1, $\sigma = 0$: linear utility, risk-neutral agents

Then (4) and (5) become identical so that in both of the cases,

$$i_t = \log \beta$$

This is the case because risk-neutral households only consider mean(first moment) for their decision. Thus, if households are risk-neutral, there is no approximation error.

Case 2, $\sigma = 5$: non-linear utility, risk-neutral agents

Using the given values and the equation (4), one can see that if we do not log-linearize, the equilibrium interest rate should be:

$$i_t \in [-\log \beta - .00125, -\log \beta - .125]$$

while $i_t = -\log \beta$ when we consider the linearized model. As a result, the approximation error is from .125% to 12.5%. Thus as increase of uncertainty (s^2), the approximation error also increases. In summary, log-linearization overestimates the interest rate (equivalently, underestimate the price of the bond) because it ignores the presence of uncertainty, and the associated precautionary savings motive, which is a driving force to drive down the interest rate (or increase the price of the bond).